Posted on 15 Aug 2024 | The copyright holder is the author/funder. All rights reserved. No reuse without permission. | https://doi.org/10.22541/au.172374899.92409871/v1 | This is a preprint and has not been peer-reviewed. Data may be preliminary.

Exploring artificial neural networks for the curation of DNA barcode reference libraries: a pilot study with marine fish species of the Eastern Atlantic

João Tadeu Fontes[1], André Oliveira Ferreira[1], Ana Sofia Lavrador[1], Cláudia Machado[1], Pedro Soares[1], Filipe Oliveira Costa[1], and Carlos Alberto Canchaya[1]

[1]Affiliation not available

August 15, 2024

***João T. Fontes[a,b] (jtadeusfontes@gmail.com), André O. Ferreira[a,b], Ana S. Lavrador[a,b] , Cláudia Machado[a,b], Pedro Soares[a,b], *Filipe O. Costa[a,b](fcosta@bio.uminho.pt), Carlos A. Canchaya[c,d]**

[a]*Centre of Molecular and Environmental Biology (CBMA) / ARNET-Aquatic Research Network, Department of Biology, University of Minho, Braga, Portugal*

[b]*Institute of Science and Innovation for Bio-Sustainability (IB-S), University of Minho, Braga, Portugal*

[c]*Department of Biochemistry, Genetics, and Immunology, University of Vigo, Vigo, Spain*

[d]*Marine Research Centre (CIM-UVIGO), Universidade de Vigo, Vigo, Spain*

**Corresponding authors*

Abstract

Accurate species assignments using (e)DNA (meta)barcoding depend on the taxonomic reliability of DNA barcode reference libraries. Human inspection of Barcode Index Numbers (BINs), a proxy for species boundaries implemented in the Barcode of Life Data System (BOLD), may flag potential discordances, and help solve pseudo-discordances. However, using this approach to process large reference libraries is labor-intensive. We explored the utility of artificial neural networks (ANNs) for automating the sorting of putatively reliable and unreliable records. We tested this approach using marine fishes, a group with relatively common taxonomic ambiguities and BIN discordances. We mined records from BOLD and extracted data features based on taxonomic identification, congruency between morphospecies and BINs, and relevant metadata. We assessed all records from species names assigned to discordant BINs, classifying them as 'reliable' or 'unreliable' using the features as a heuristic guide. For a dataset with 63,302 records, comprising 3,055 species names and 1,865 BINs, we used a fivefold random subsampling validation approach, allocating 1,400 BINs for training and 465 for testing in each iteration. Both training and validation displayed tightly aligned accuracy and loss, decreasing the risk of overfitting. Across iterations, ANNs achieved predictive accuracies between 91.4% and 96.3%. This performance highlights the models' effectiveness in replicating manual curation, significantly reducing the time and effort required. This pilot study provides encouraging results for applying ANNs to the curation of reference libraries, suggesting a promising direction for future research, where deep learning could facilitate or enhance the meticulous work of taxonomists and data curators.

Posted on 15 Aug 2024 | The copyright holder is the author/funder. All rights reserved. No reuse without permission. | https://doi.org/10.22541/au.172374899.92409871/v1 | This is a preprint and has not been peer-reviewed. Data may be preliminary.

Keywords: marine fish, DNA barcoding, DNA metabarcoding, eDNA, machine learning, artificial intelligence, auditing, species-level identification

Introduction

Reference libraries are the basis for taxonomic identification through (e)DNA (meta)barcoding approaches, making their accuracy crucial for incorporating these molecular identification tools into biodiversity surveys, conservation, or fisheries management. Misidentifications and ambiguities commonly emerge within these libraries and can compromise the accuracy of taxonomic assignments (Cheng *et al.* , 2023; Keck*et al.* , 2023; Packer *et al.* , 2009; Pentinsaari *et al.* , 2020). This is especially the case with (e)DNA metabarcoding studies, where a high volume of sequencing reads is often queried in an automated way against reference sequence data (Bengtsson-Palme *et al.* , 2018; Kirse *et al.* , 2021; Lavrador *et al.* , 2023). Even though efforts have been made to diminish the impact of errors and incongruencies (Antil *et al.* , 2023; Claver et a., 2023; Fontes*et al.* , 2021; Ershova *et al.* , 2023; Gostel & Kress, 2022), ultimately, the problem of taxonomic misidentifications inadvertently entering the reference sequence data generation workflow and propagating through DNA-based biodiversity studies remains largely unsolved.

Although there is no flawless approach for identifying errors and ambiguities, several useful systems have been developed to compile and curate reference libraries. The software package *crabs* (Jeunen*et al.* , 2023) incorporates multiple data quality control measures, including the identification of records with incomplete or ambiguous taxonomic information, though it does not pinpoint potential errors or misidentifications. The R packages *rCRUX* (Curd *et al.* , 2024) and *refdb* (Keck & Altermatt, 2023) are notable for their robust data mining and quality control features. For instance, they can detect sequences that are identical but have different upstream taxonomies, effectively determining ambiguities in the records' taxonomic identification, although they do not address taxonomic congruency between morphospecies and Molecular Operational Taxonomic Units (MOTUs). Another system, mkCOInr (Meglécz, 2023), is highly effective at dereplicating barcode data from BOLD and GenBank. It includes steps to avoid ambiguous taxonomy and synonyms, although it does not assess taxonomic congruency or identify potential erroneous records.

The Barcode Index Number (BIN) system (Ratnasingham & Hebert, 2013) implemented in the Barcode of Life Data system (BOLD; Ratnasingham & Hebert, 2007), is central to the process of guaranteeing the reliability of DNA barcode reference libraries. This system clusters COI barcode sequences in a way that, ideally, mirrors the delimitation of morphospecies. It acts as a tool for organizing and verifying species identities by grouping sequences that exhibit a high degree of genetic similarity, ensuring that each species is represented by a consistent and specific MOTU. However, the occurrence of species names assigned to discordant BINs – BINs comprising more than one species name – is frequent for several taxonomic groups, including fishes, both due to operational and biological reasons (Fontes *et al.* , 2021; Lavrador*et al.* , 2021; Radulovici *et al.* , 2021). Although BOLD offers users a tool to highlight and further examine these BINs given a set of available records, the curation of these BINs remains unresolved.

Costa *et al.* (2012) developed and tested a morphospecies-centered qualitative grade system to assess the congruency between morphospecies and sequence clusters, using marine fishes from Portugal as a case study. Oliveira *et al.* (2016) updated this system to use BINs as MOTUs and audited a reference library for European marine fish species using this approach. Later, Fontes *et al.* (2021) automated the process by developing BAGS (Barcode, Audit & Grade System), an R-based system that mines public records from BOLD and assigns grades according to the congruency between species names and BINs at a given point in time. Although this system is capable of pinpointing concordant correspondences between morphospecies and BINs, as well as species names that are assigned to at least one discordant BIN, it does not differentiate between pseudo-discordances (i.e., two synonym species names sharing a BIN) and actual incongruencies, such as potential morphological misidentifications.

Thus, there is a need to curate discordant BINs by pinpointing records potentially arising from operational errors and detecting cases where biological phenomena may aggregate several morphospecies in a single MOTU. This is especially the case when the research aims for species-level identification, where the ideal scenario

Posted on 15 Aug 2024 | The copyright holder is the author/funder. All rights reserved. No reuse without permission. | https://doi.org/10.22541/au.172374899.92409871/v1 | This is a preprint and has not been peer-reviewed. Data may be preliminary.

is for each BIN to correspond to a single species name. To achieve this level of data curation, researchers usually need to manually inspect the records' data and metadata to sort out potential misidentifications and segregate reliable and unreliable barcodes (Ferreira *et al.* , 2024; Lavrador *et al.* , 2023; Magoga *et al.* , 2022; Nilsson *et al.* , 2018; Radulovici *et al.* , 2021). However, this approach is rather time-consuming and labor-intensive, especially for larger datasets. In this context, artificial neural networks (ANNs) may represent a promising solution for automating and streamlining the process of curating large volumes of data. ANNs are already being utilized to automate other labor-intensive tasks that traditionally require human expertise and decision-making, such as medical diagnosis (Raghavendra *et al.* , 2020) and speech recognition (Dhanjal & Singh, 2024).

In this pilot study, we tested the application of ANNs to automate and expedite the process of manual BIN curation (Figure 1). For this purpose, we used a DNA barcode reference library comprising Eastern Atlantic marine fish species. This group has been increasingly targeted in (e)DNA (meta)barcoding studies (Haldar & Nath, 2020; Xiong *et al.* , 2022), where taxonomic ambiguities and BIN discordances are relatively common (Claver *et al.* , 2023; Fontes *et al.* , 2021; Oliveira *et al.* , 2016). Moreover, since species-level identification is crucial for fisheries management and stock assessments, the successful incorporation of DNA-based approaches into this field depends on maintaining an exceptionally high standard of accuracy in DNA barcode reference libraries. Thus, resolving situations where fish species names are assigned to discordant BINs is extremely important. Our main objective was to understand whether deep learning models are capable of mirroring human assessment when sorting out putatively reliable and unreliable DNA barcode records, hopefully laying the foundation for further application of artificial intelligence in this endeavor.

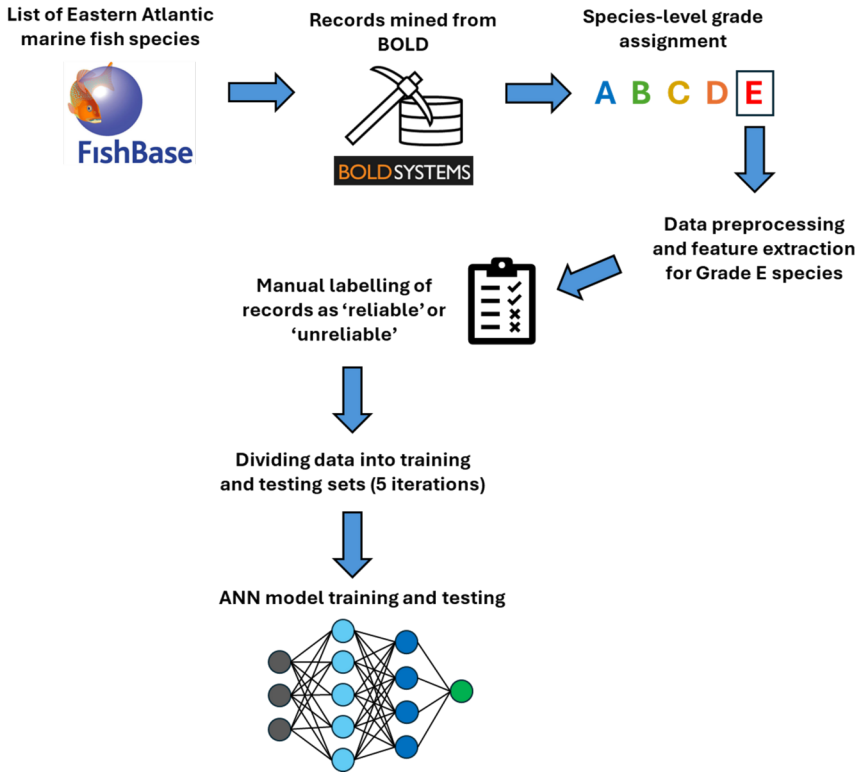


Figure 1: Figure 1 - Overall view of the workflow utilized for testing the use of ANNs in the curation of species names assigned to discordant BINs.

Methodology

Posted on 15 Aug 2024 | The copyright holder is the author/funder. All rights reserved. No reuse without permission. | https://doi.org/10.22541/au.172374899.92409871/v1 | This is a preprint and has not been peer-reviewed. Data may be preliminary.

Data mining

We assembled a list of fish species occurring in the Eastern Atlantic Ocean and connected bodies of water from FishBase (Froese & Pauly, 2023), resorting to the *rfishbase* R package (Boettiger *et al.* , 2012) in December 2023 and removed species annotated with "error", "misidentification" and "questionable". Subsequently, we exclusively retained marine species and standardized the taxonomic information using the World Register of Marine Species (WoRMS Editorial Board, 2023). Using the assembled species list as a query, we adapted the code from BAGS (Barcode, Audit & Grade System; Fontes *et al.* , 2021) to mine and annotate the respective DNA barcode records from BOLD. Contrary to the standard BAGS pipeline, since we aimed to include all records, regardless of sequence quality, we did not apply any filtering steps except for removing records without species-level annotation or assigned BINs. To ensure comprehensiveness, we further mined all records from species clustered in BINs assigned to the previously obtained records of Eastern Atlantic marine fish species, regardless of the geographical distribution of these additional species, as BINs serve are unit of taxonomic congruency applied in this study.

Data preprocessing and feature extraction

Most of the data preprocessing and feature extraction steps were conducted using the R programming language, within the integrated environment of RStudio (Allaire, 2012) version 2021.9.1. Qualitative grades, from A to E, were assigned to each species in the dataset according to the congruency between species names and sequences clustered in BINs, as described by Fontes *et al.* (2021). Grades A and B comprised species names assigned to a single BIN integrating solely records with that species name; grade C comprised species names assigned to more than one BIN, albeit integrating only records from said species; grade D consisted of non-discordant species names (i.e., non-grade E) with fewer than three records in the data set; and grade E consisted of species names assigned to at least one BIN comprising more than one species name (i.e., the species name may occur in several BINs, but at least one of these BINs contains additional species names). Considering that our goal was to further identify the reliable and unreliable records, the subsequent analyses focused exclusively on grade E records.

For each record in the grade E dataset associated with a specific species name clustered in a BIN, we extracted categorical, numerical, and binary features to assess whether the record was putatively reliable or unreliable. Categorical features consisted of the species name, the genus, and the BIN. Numerical features included: the number of distinct species names within the BIN, the number of distinct BINs assigned to the species name, the total number of records for the species names in the dataset, the total number of records in the respective BIN, the number of records for that particular species name in the BIN, and the percentage the latter represented of the total records within the BIN. Additionally, we included the number of different individuals who identified at least one specimen of the species clustered in the BIN, as well as the number of distinct institutions housing vouchers of the species recorded in the BIN. For species names with no information on taxonomic identifier or housing institution in a particular BIN, the value was zero.

To further refine our reliability assessment, we also determined two binary features: the first informed whether the species shared the BIN with at least one synonym species name, which was achieved through the WoRMS database resorting to the *taxize* R package (Chamberlain & Szöcs, 2013); and the second checked whether the species name contains ambiguous expressions (e.g., sp., cf., complex., group., etc.) or simply does not follow the binomial nomenclature. To ensure compatibility with the neural network models, categorical variables were encoded via label encoding. Additionally, the percentage feature underwent standardization, resulting in a mean of zero and a standard deviation of one, while the remaining numerical features were log-transformed for normalization.

Dataset labeling with ground truths

During the process of labeling the data with ground truths, we utilized the features mainly as a heuristic guide rather than exclusively establishing thresholds through a strictly rule-based system (see Table S1). This approach was chosen to ensure that we could test whether the neural network was able to mimic the nuanced decision-making involved in curating barcodes from species assigned to discordant BINs. Each

Posted on 15 Aug 2024 | The copyright holder is the author/funder. All rights reserved. No reuse without permission. | https://doi.org/10.22541/au.172374899.92409871/v1 | This is a preprint and has not been peer-reviewed. Data may be preliminary.

record was categorized as 'reliable' when the accumulated evidence and context from the features suggested that the record was correctly identified. In contrast, a record was labeled as 'unreliable' when potential misidentifications were detected, when multiple consolidated morphospecies were grouped in one BIN, or if the record's reliability remained inconclusive.

For instance, if a BIN contained 95 records for species A, with the taxonomic identifications deriving from ten individuals; and five records for species B, identified by only one individual, then the records for species A would be labeled as 'reliable', while species B records would be labeled as 'unreliable'. This rationale is based on the majority rule outlined by Ratnasingham and Hebert (2013), which considers both the predominant number of records and the agreement among multiple taxonomists when determining the reliability of identifications. On the other hand, if a BIN featured 50 records from species A and 50 records from species B, all records would be considered 'unreliable', except in cases where the species names are synonyms. In that case, all records in the BIN would be labeled as 'reliable' since both species names represent the same morphospecies.

In addition, all features aided in determining the reliability of the records, increasing, or decreasing the probability of a record's classification as 'reliable' or 'unreliable'. While the majority rule played a significant role in assessing the records' reliability, the proportion of records from each species was evaluated in conjunction with all the remaining features. The only two rule-based criteria we applied were the following: 1) if a BIN contained only one species name, the records were labeled as 'reliable' solely if the BIN contained three or more records (i.e., to maintain the same reasoning behind grade D assignment; Fontes *et al.* , 2021); 2) all records with ambiguous species names were considered 'unreliable' as they did not provide unambiguous species-level identification.

Artificial neural network implementation

The neural network models created for this study were implemented in Python version 3.11.5 within the Spyder IDE (Raybaut, 2009), utilizing Keras as a high-level API (Kapoor *et al.* , 2022). To ensure robustness and reliability in the evaluation of our models, we adopted a subsampling validation approach by splitting the labelled grade E dataset into training and testing sets across five different data splits. In each iteration, the dataset was randomly divided into training and testing sets, with 1400 BINs allocated for training and 465 bins allocated for testing.

For each split, a feedforward neural network model was structured using the 'Sequential' class with fully connected neurons, including one input layer, two hidden layers and an output layer. The input layer included 13 neurons, one for each of the aforementioned features while the first hidden layer included 64 neurons and the second 32 neurons. The hidden layers were equipped with rectified linear unit (ReLU) activation functions (Nair & Hinton, 2010), enabling the introduction of non-linearity and the extraction of patterns from the data. Additionally, dropout regularization, with a dropout rate of 5%, was applied after the first hidden layer to mitigate potential overfitting issues and enhance the model's generalizability. The output layer consisted of a single neuron with a sigmoid activation function, allowing the model to produce binary predictions, making it suitable for classification into positives and negatives, corresponding to the 'reliable' and 'unreliable' classes, respectively.

During the training phase, the model parameters were optimized using the Adam optimizer (Kingma & Ba, 2014) in conjunction with a binary cross-entropy loss function. Training occurred over 50 epochs, with a batch size of 64 instances per batch. We used a validation split comprising 10% of the training data, enabling the monitoring of the models' performance on unseen data, ensuring its generalizability, and further avoiding overfitting. The architecture and hyperparameters of the models were selected after thorough experimentation with different configurations. Following training, each model's performance and predictive capability were assessed using the independent testing dataset previously unseen during training. Several evaluation metrics were calculated, namely the accuracy (proportion of correct predictions among all predictions), loss (difference between the model's predictions and the actual labels), precision (ratio of correctly predicted positive/negative observations to the total number of predicted positives/negatives), recall (ratio of correctly

Posted on 15 Aug 2024 | The copyright holder is the author/funder. All rights reserved. No reuse without permission. | https://doi.org/10.22541/au.172374899.92409871/v1 | This is a preprint and has not been peer-reviewed. Data may be preliminary.

predicted positive/negative observations to all observations in the actual class), and F1-score (the weighted average of precision and recall). The R and Python scripts used for data mining, preprocessing, feature extraction and ANN implementation, as well as the ANN models described in this study and their associated files and datasets, are available at https://github.com/tadeu95/ANNs-DNA-Barcode-Curation.

Results

The mined dataset for Eastern Atlantic marine fishes included 83,473 records, comprising 3,991 unique species names, 2,956 accepted species names (i.e., after removal of ambiguous species names) and 3,072 BINs. The grade E species dataset included 63,302 records, 3,055 species names, 2,020 accepted species names, and 1,865 BINs. The distribution of the qualitative grades can be observed in Figure 2, with grade E species comprising approximately 75% of the total number of species names, and approximately 68% of the accepted species names. The manually labeled grade E dataset consisted of 36,308 records labeled as 'reliable', and 26,994 records labeled as 'unreliable'. The composition of the five different data splits for the labeled grade E data set is displayed in Table 1. Across the datasets, the size of the training set ranged from 75% to 79.4% of the complete grade E species dataset, while the testing set ranged from 20.6% to 25%.

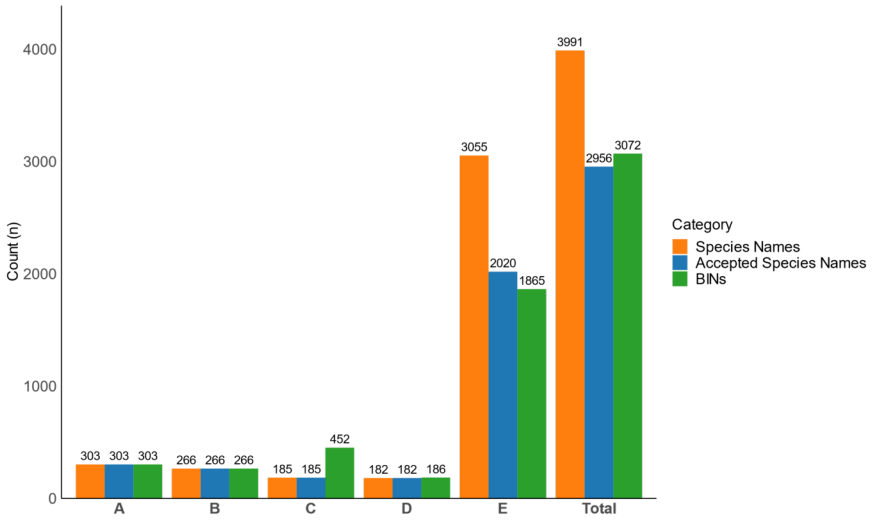


Figure 2: Figure 2 - Distribution of the number of species names, accepted species names, and BINs in the dataset assigned through the BAGS auditing system to each qualitative grade (A: consolidated concordance; B: basal concordance; C: multiple BINs; D: insufficient data; E: discordant species assignment).

Table 1 - Summary of training and testing set composition across five data splits.

Dataset	Records	Species	BINs	% records	
Split 1	Training	49,544	2,508	1,400	78.3%
	Testing	13,758	918	465	21.7%
Split 2	Training	48,575	2,510	1,400	76.7%
	Testing	14,727	897	465	23.3%
Split 3	Training	50,275	2,542	1,400	79.4%
	Testing	13,027	871	465	20.6%
Split 4	Training	50,037	2,539	1,400	79.0%
	Testing	13,265	888	465	21.0%
Split 5	Training	47,463	2,428	1,400	75.0%

Posted on 15 Aug 2024 | The copyright holder is the author/funder. All rights reserved. No reuse without permission. | https://doi.org/10.22541/au.172374899.92409871/v1 | This is a preprint and has not been peer-reviewed. Data may be preliminary.

Dataset	Records	Species	BINs	% records
	Testing	15,839	1,021	465	25.0%

Figure 3 illustrates the average training and validation accuracy, as well as the loss, of the models across five random data splits over 50 training epochs. Both accuracy plots show an upward trend, with training and validation accuracies closely tracking each other and plateauing toward the end of training, suggesting minimal risk of overfitting. Similarly, the loss plots decrease over time, with training and validation losses converging to low values, indicating consistent model performance and generalizability.

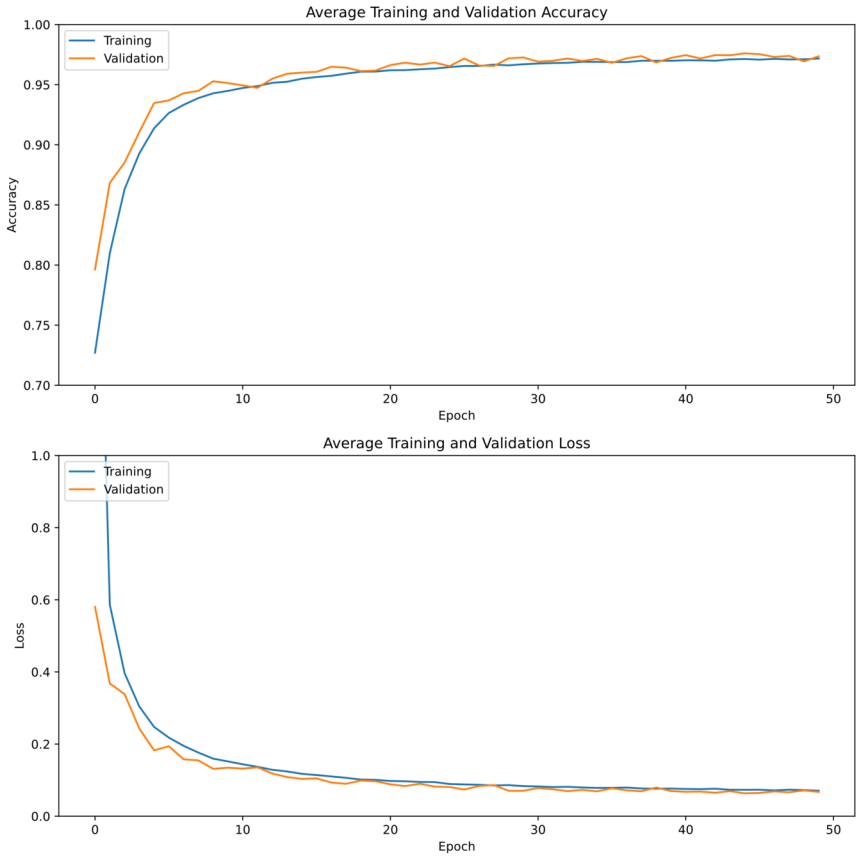


Figure 3: Figure 3 - Average training and validation accuracy and loss for the five models over 50 training epochs. Averages are computed from five random data splits.

Each model took approximately one to two minutes to train and to test on unseen data using personal computers. The predictive accuracy ranged from 91.4% for split 4 to 96.3% for split 3, with an average of 93.5% correct predictions across the five splits (Table 2). Both the precision and F1 score were greater for the 'reliable' class in four out of the five splits, with split 5 being the exception in both cases. On the other hand, recall was also higher for the 'reliable' class in four of the splits, although the exception in this case was split 3. The confusion matrix, displaying the percentages of true positives, true negatives, false positives, and false negatives for the five models, is represented in Figure 4. Overall, false negatives remained consistent across all splits, ranging from 3.0% in split 1 to 4.0% in split 4. In contrast, false positives demonstrated greater variability, ranging from 0.6% of predictions in split 3 to 4.8% in split 5.

Posted on 15 Aug 2024 | The copyright holder is the author/funder. All rights reserved. No reuse without permission. | https://doi.org/10.22541/au.172374899.92409871/v1 | This is a preprint and has not been peer-reviewed. Data may be preliminary.

Table 2 – Prediction performance metrics across the five data splits.

	Label	Precision	Recall	F1-score	Support	Accuracy
Split1	Unreliable Reliable	91.5% 96.7%	93.8% 95.5%	92.6% 96.1%	4,697 9,061	94.9%
Split2	Unreliable Reliable	90.5% 93.7%	88.4% 94.9%	89.4% 94.3%	5,224 9,503	92.6%
Split3	Unreliable Reliable	93.6% 98.9%	98.7% 94.2%	96.0% 96.5%	5,969 7,058	96.3%
Split4	Unreliable Reliable	90.7% 91.9%	89.7% 93.0%	90.2% 92.4%	5,841 7,424	91.4%
Split5	Unreliable Reliable	94.5% 90.5%	90.5% 94.5%	92.5% 92.4%	8,102 7,737	92.5%
Average	Unreliable Reliable	92.2% 94.3%	92.2% 94.4%	92.1% 94.3%	5966.6 8156.6	93.5%

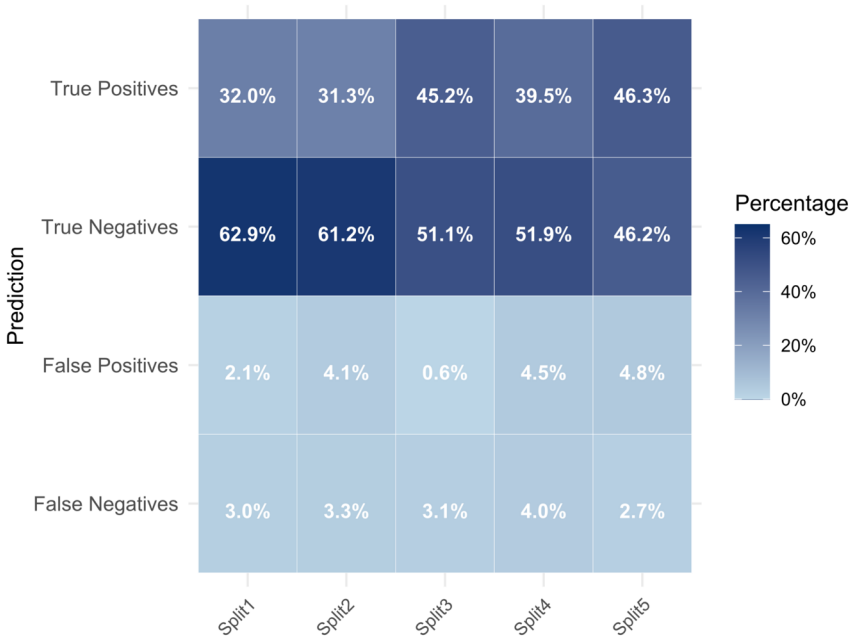


Figure 4: Figure 4 - Confusion matrix displaying the percentage of true positives, true negatives, false positives, and false negatives for five models across data splits (split1 to split5).

Discussion

With the continuous expansion of DNA barcode reference libraries, the increase in their volume not only enhances their utility but also heightens the risk of errors and inconsistencies, which can significantly impact the accuracy of taxonomic assignments. Currently, there are over 17.8 million barcodes in BOLD (accessed on the 13th of July 2024). As the database grows and new barcodes are assigned to BINs, managing these data becomes increasingly complex, and the emergence of incongruencies is unavoidable. In the case of fishes, where species-level identification is often the main goal in (e)DNA (meta)barcoding studies (Cicala *et al.* , 2024; Duhamet*et al.* , 2023), employing BINs as proxies for morphospecies is valid only if a conclusive determination can be made regarding the actual morphospecies that corresponds to each BIN.

The mined COI dataset for Eastern Atlantic fishes comprised a high proportion of species names assigned to at least one discordant BIN, with approximately 68% of accepted morphospecies names assigned to grade E (Figure 2). However, as it has been previously reported (Fontes*et al.* , 2021; Oliveira *et al.* , 2016), this number can be drastically reduced by the detection of pseudo-discordances, mainly caused by ambiguous

Posted on 15 Aug 2024 | The copyright holder is the author/funder. All rights reserved. No reuse without permission. | https://doi.org/10.22541/au.172374899.92409871/v1 | This is a preprint and has not been peer-reviewed. Data may be preliminary.

species names, synonyms and morphospecies grouped in the same BIN. Therefore, since grade E assignment is merely a way to pinpoint species names that need a closer inspection of their records – rather than ascertaining the records' unreliability – these cases were the focus of this study.

We opted for a heuristic approach (i.e., not establishing a fixed rule-based system) where the records' data and metadata, as well as newly extracted features, serve as a basis for segregating the most trustworthy records, detecting pseudo-discordances, and pinpointing potential errors. This approach highlights the reliance on manual BIN inspection, which, despite its limitations (i.e., there is no present way of determining a record's reliability with absolute certainty), remains one of the only methods currently available for resolving the ambiguities associated with discordant BINs (Ferreira *et al.* , 2024; Lavrador *et al.* , 2023; Magoga *et al.* , 2022; Radulovici *et al.* , 2021).

In addition to the increase in the volume of barcode data, because of the labor-intensive nature of manual BIN curation, ANNs are a promising resource for the future of curating reference libraries due to their ability to recognize patterns in data and mimic human decision-making (Goodfellow *et al.* , 2016). Indeed, deep learning has begun to be utilized for various purposes in the field of molecular ecology, such as species classification and biodiversity assessment (Lamperti *et al.* , 2023; Müller *et al.* , 2023; Yang *et al.* , 2021). The findings of this pilot study confirm the potential of ANNs in automating the curation of DNA barcodes, as evidenced by the robust generalization of the models. This was demonstrated by the training and validation accuracy and loss plots (Figure 3), the low proportion of false positives and negatives (Figure 4), and their consistent ability to predict, on average, 93.5% of all labels across data splits (Table 2).

We chose to divide the labeled dataset into training and testing sets based on the number of BINs (i.e., 1,400 BINs for training and 465 BINs for testing, Table 1), rather than on the number of records or species names, since BINs represent the basic unit of taxonomic congruency in this study. This division allowed ANNs to base their decision on the information for all records contained within a particular BIN, making it ideal for the training phase. Moreover, each record included features related to other BINs in the dataset (i.e., number of species names per BIN and number of BINs per species name), thereby increasing the information available both for manual curation and for the ANN's pattern recognition. Overall, the prediction metrics for both the 'reliable' and 'unreliable' classes (Table 2) demonstrated a robust performance of ANNs for this application, particularly given that this was a pilot study. Furthermore, this approach can be easily adjusted to accommodate other taxonomic groups, as well as alternative MOTU systems, solely requiring some modifications to the workflow and code.

ANNs could predict a high proportion of pseudo-discordances, where the erroneous nature of the records is merely apparent. For instance, in the predictions made by the split 2 model, all records annotated with the two synonym species names present in the BIN BOLD:ACG9497, *Apogon smithi* and *Jaydia smithi* , were classified as 'reliable' since both species names represent the same morphospecies. The ANN models were also capable of matching the ground truth labels in other scenarios of synonymy, where putative erroneous records are present. In the split 1 model, the testing dataset included BIN BOLD:AAB6180, which comprised three species names, the first two being synonyms: *Sardinops melanosticta* , *Sardinops sagax* , and *Spratelloides robustus.* Since *Spratelloides robustus* comprised only one record (making up 1.5% of the records in the BIN) and a potential misidentification, the ANN classified it as 'unreliable' while classifying the remaining records in the BIN as 'reliable'.

Pseudo-discordances occurring due to the presence of ambiguous species names were also mostly detected by the ANNs. For example, in the split 4 model, BIN BOLD:ACK8065 comprised two species names, *Chirocentrus dorab* , which was identified by five different individuals in the BIN, and *Chirocentrus cf. dorab* , indicating a species identification that has not been fully confirmed. The ANN replicated our ground truth labeling by classifying the *C. dorab* records as 'reliable', and the *C. cf. dorab records* as 'unreliable' because the latter do not comprise confirmed species-level identifications. Similarly, in the split 1 model, BOLD:ACF1143 included mostly *Gadus morhua* records, classified as 'reliable' by the ANN, with a single *Gadus sp.* record, which was classified as 'unreliable' for not providing species-level identification.

Posted on 15 Aug 2024 | The copyright holder is the author/funder. All rights reserved. No reuse without permission. | https://doi.org/10.22541/au.172374899.92409871/v1 | This is a preprint and has not been peer-reviewed. Data may be preliminary.

While false negatives comprised a small proportion of the classifications made by the ANN models (Figure 4), we did observe a few instances, specifically in scenarios of pseudo-discordances. For instance, in split 5, BOLD:AAE9134 comprised two synonym species names, *Planiliza haematocheila*, which made up 78.9% of records in the BIN and was identified by four different individuals, and *Planiliza haematocheilus,* which did not have information regarding the taxonomist who identified the specimens. The ANN model applied the majority rule (Ratnasingham & Hebert, 2013) by classifying only the *P. haematocheila* records as 'reliable'. This is most likely because, although pseudo-discordances due to synonyms are relatively common for marine fishes, they represent a small proportion of the patterns found in the current dataset, with only 82 species names (i.e., out of a total of 3991 species names) having a synonym name within their respective BIN. Consequently, not all of the five models were trained with the same proportion of cases of synonymy, limiting their ability to generalize these scenarios with unseen data. Therefore, in future applications, increasing the training dataset will likely make the occurrence of these patterns more frequent, thereby improving the model's ability to correctly classify such cases.

Additionally, although all records with ambiguous species names were classified as 'unreliable' by the ANNs, there were occurrences of false negatives in BINs where this type of pseudo-discordance were present. For example, BIN BOLD:AAB1178 included, in addition to records belonging to *Chauliodus sloani (identified by 15 different individuals)* , 85 other species names that consisted of *Chauliodus sloani sp.* followed by different alphanumerical identifiers (e.g., *Chauliodus sp. DPND_ 1895*). During manual labeling, although we labeled the*C. sloani* records as 'reliable', the ANN classified all records in this BIN as 'unreliable'. This is most likely due to the high number of distinct ambiguous species names, which the ANN might have confused with scenarios of multiple species names aggregated in a single BIN.

One example of multiple species names lumping in one BIN is BOLD:AAA7352, which included records from eight different morphospecies names of the *Thunnus* genus, a group where interspecific genetic distances for DNA barcodes can be very low (Lowenstein *et al.* , 2009; Ward *et al.* , 2005). In this and other similar scenarios, the ANN classification of all the BINs' records generally matched our manual labeling, classifying them as 'unreliable'. In these cases, the aggregation of morphospecies is a pseudo-discordance as it does not necessarily reflect operational incongruencies or misidentifications, but instead shows an inability of the BIN system and the COI variation to discriminate those morphospecies. This prevents a precise match between morphospecies and BIN, leading to the classification of the records as 'unreliable'. This lack of precise matching can lead to significant errors, especially when automatically processing metabarcoding data.

ANNs were efficient at recognizing cases where putative morphological misidentifications or operational errors were present. For example, in the split 1 model, BOLD:AAA5956 consisted of 411 *Scomber scombrus* records identified by 13 different individuals, and 2 *Somber japonicus* records without information regarding the identifier. We labeled the *S. scombrus* records as 'reliable' and the *S. japonicus* records as 'unreliable', which was mimicked by the ANN. Another instance is BOLD:ADF5681, which included records for three morphospecies: *Lutjanus fulviflamma* , *Lutjanus johnii,* and*Lutjanus monostigma* . *L. fulviflamma* was identified by 17 individuals, solely assigned to BOLD:ADF5681, and comprised 86.6% of records in the BIN. On the other hand, *L. johnii* was assigned to one additional BIN and was identified by only one individual. Finally,*L. monostigma* was assigned to two additional BINs and did not have information regarding the identifier. Therefore, the ANN mimicked the ground truth labeling by classifying *L. fulviflamma* records as 'reliable' and the remaining as 'unreliable'.

Although false positives comprised an overall small percentage of all classifications (Figure 4) there were instances where the uncertainty inherent to BIN curation was reflected in the models' classifications. For example, BOLD:AAA1547 included two closely related morphospecies (Ebert *et al.* , 2010), *Squalus acanthias* and *Squalus suckleyi* , the former making up 86.9% of records in the BIN and identified by seven different individuals, and the latter making up 13.1% of records, albeit identified by five individuals. Despite*S. acanthias* comprising most of the records in this BIN and having been identified by seven different individuals, we labeled all records in this BIN 'unreliable'. This was because, overall, the features suggested that the discrimination of these two species might be challenging. However, the split 4 model classified the *S.*

acanthias records as 'reliable', and the *S. suckleyi* records as 'unreliable'. This was a common trend among misclassifications across data splits, with ANN models tending to be generally riskier in their decision-making by applying the majority rule (Ratnasingham & Hebert, 2013).

Conclusion

The curation of continuously expanding reference sequence libraries is, currently, an unresolved and ongoing challenge in the scientific community. Several auditing and curation workflows and tools have been implemented to increase the quality of reference libraries, both within the BOLD environment and through systems from external sources. Nevertheless, the complexity and sheer volume of data outpace the current capabilities of these systems, with researchers resorting to manual data curation to complement the auditing of reference libraries. As evidenced by the performance of our pilot implementation, ANNs have the potential to accelerate these curation endeavors. In the future, increasing the size of training datasets and expanding them for other taxonomic groups is essential for improving the accuracy and applicability of the models. Indeed, a potential way to increase the pattern recognition capabilities of ANN models could be by combining training labeled datasets from different taxonomic groups, especially if similar patterns are found between them. Moreover, including additional data features and integrating the information from multiple databases could help improve the manual curation process and the ANN's efficiency. Overall, the advent of artificial intelligence will certainly influence the way molecular identification tools are implemented, applied, and validated, thereby promising significant advancements for future applications.

Contributions

J.T.F., P.S., F.O.C., and C.A.C. designed and conceived the research. J.T.F, A.O.F, A.S.L, and C.M. performed the manual data curation and labeling. J.T.F wrote the programming code and performed data analysis with the help of C.A.C. J.T.F. wrote the manuscript with the help of all authors. All authors contributed to the manuscript and approved the final version.

Data availability statement

The programming scripts used in this study are publicly available on GitHub (https://github.com/tadeu95/ANNs-DNA-Barcode-Curation). Additionally, the data sets, trained models, and associated files utilized in this study are also available on the GitHub repository.

Hosted file

`Table S1.xlsx` available at https://authorea.com/users/317355/articles/1215586-exploring-artificial-neural-networks-for-the-curation-of-dna-barcode-reference-libraries-a-pilot-study-with-marine-fish-species-of-the-eastern-atlantic

Conflict of interest. The authors declare that they do not have any conflict of interest.

Funding

This work was funded by the project "A-FISH-DNA-SCAN: Cutting-edge DNA-based approaches for improved monitoring and management of fisheries resources along Magellan-Elcano's Atlantic route" (http://doi.org/10.54499/CIRCNA/BRB/0156/2019), funded by the Portuguese foundation of Science and Technology (FCT, I.P. reference: CIRCNA/BRB/0156/2019), and by the "Contrato-Programa" (https://doi.org/10.54499/UIDB/04050/2020), funded by national funds through the FCT I.P. J.T.F. (UI/BD/150910/2021; https://doi.org/10.54499/UI/BD/150910/2021) and A.S.L. (UI/BD/150871/2021; https://doi.org/10.54499/UI/BD/150871/2021) are supported by the Collaboration Protocol for Financing the Multiannual Research Grants Plan for Doctoral Students with financial support from FCT I.P. and the European Social Fund under the Northern Regional Operational Program — Norte2020. A.O.F. was supported by grant DFA/BD/6653/2020 (https://doi.org/10.54499/2020.06653.BD) from FCT I.P.

References

Posted on 15 Aug 2024 | The copyright holder is the author/funder. All rights reserved. No reuse without permission. | https://doi.org/10.22541/au.172374899.92409871/v1 | This is a preprint and has not been peer-reviewed. Data may be preliminary.

Posted on 15 Aug 2024 | The copyright holder is the author/funder. All rights reserved. No reuse without permission. | https://doi.org/10.22541/au.172374899.92409871/v1 | This is a preprint and has not been peer-reviewed. Data may be preliminary.

Antil, S., Abraham, J. S., Sripoorna, S., Maurya, S., Dagar, J., Makhija, S., Bhagat, P., Gupta, R., Sood, U., Lal, R., & Toteja, R. (2023). DNA barcoding, an effective tool for species identification: a review. *Molecular biology reports, 50* (1), 761-775. https://doi.org/10.1007/s11033-022-08015-7

Bengtsson-Palme, J., Richardson, R. T., Meola, M., Wurzbacher, C., Tremblay, É. D., Thorell, K., Kanger, K., Eriksson, K. M., Bilodeau, G. J., Johnson, R. M., Hartmann, M., & Nilsson, R. H. (2018). Metaxa2 Database Builder: Enabling taxonomic identification from metagenomic or metabarcoding data using any genetic marker. *Bioinformatics ,34* (23), 4027–4033. https://doi.org/10.1093/bioinformatics/bty482

Boettiger, C., Lang, D. T., & Wainwright, P. C. (2012). rfishbase: Exploring, manipulating and visualizing FishBase data from R.*Journal of Fish Biology* , *81* (6), 2030–2039. https://doi.org/10.1111/j.1095-8649.2012.03464.x

Chamberlain, S. A., & Szöcs, E. (2013). taxize: taxonomic search and retrieval in R. *F1000Research* , *2*. https://doi.org/10.12688/f1000research.2-191.v2

Cheng, Z., Li, Q., Deng, J., Liu, Q., & Huang, X. (2023). The devil is in the details: Problems in DNA barcoding practices indicated by systematic evaluation of insect barcodes. *Frontiers in Ecology and Evolution, 11* , 1149839. https://doi.org/10.3389/fevo.2023.1149839

Cicala, D., Maiello, G., Fiorentino, F., Garofalo, G., Massi, D., Sbrana, A., Mariani, S., D'Alessandro, S., Stefani, M., Perrodin, L., & Russo, T. (2024). Spatial analysis of demersal food webs through integration of eDNA metabarcoding with fishing activities.*Frontiers in Marine Science* , *10* . https://doi.org/10.3389/fmars.2023.1209093

Claver, C., Canals, O., de Amézaga, L. G., Mendibil, I., & Rodriguez-Ezpeleta, N. (2023). An automated workflow to assess completeness and curate GenBank for environmental DNA metabarcoding: The marine fish assemblage as case study. *Environmental DNA ,5* (4), 634–647. https://doi.org/10.1002/edn3.433

Costa, F. O., Landi, M., Martins, R., Costa, M. H., Costa, M. E., Carneiro, M., Alves, M. J., Steinke, D., & Carvalho, G. R. (2012). A ranking system for reference libraries of DNA barcodes: Application to marine fish species from Portugal. *PLoS One* , *7* (4), e35858. https://doi.org/10.1371/journal.pone.0035858

Curd, E. E., Gal, L., Gallego, R., Silliman, K., Nielsen, S., & Gold, Z. (2024). rCRUX: A rapid and versatile tool for generating metabarcoding reference libraries in R. *Environmental DNA ,6* (1), e489. https://doi.org/10.1002/edn3.489

Dhanjal, A. S., & Singh, W. (2024). A comprehensive survey on automatic speech recognition using neural networks. *Multimedia Tools and Applications* , *83* (8), 23367–23412. https://doi.org/10.1007/s11042-023-16438-y

Duhamet, A., Albouy, C., Marques, V., Manel, S., & Mouillot, D. (2023). The global depth range of marine fishes and their genetic coverage for environmental DNA metabarcoding. *Ecology and Evolution ,13* (1), e9672. https://doi.org/10.1002/ece3.9672

Ebert, D. A., White, W. T., Goldman, K. J., Compagno, L. J., Daly-Engel, T. S., & Ward, R. D. (2010). Resurrection and redescription of*Squalus suckleyi* (Girard, 1854) from the North Pacific, with comments on the *Squalus acanthias* subgroup (Squaliformes: Squalidae). *Zootaxa* , *2612* (1), 22–40. https://doi.org/10.11646/zootaxa.2612.1.2

Ershova, E. A., Wangensteen, O. S., & Falkenhaug, T. (2023). Mock samples resolve biases in diversity estimates and quantitative interpretation of zooplankton metabarcoding data. *Marine Biodiversity, 53* (5), 66. https://doi.org/10.1007/s12526-023-01372-x

Ferreira, A. O., Azevedo O. M., Barroso, C., Duarte, S., Egas, C., Fontes, J. T., Re, P., Piecho-Santos, A. M., Costa, F. O. Multi-marker DNA metabarcoding for precise species identification in ichthyoplankton samples: a proof-of-concept study. (2024). Preprint (Version 1) available at Research Square. https://doi.org/10.21203/rs.3.rs-4600568/v1

Posted on 15 Aug 2024 | The copyright holder is the author/funder. All rights reserved. No reuse without permission. | https://doi.org/10.22541/au.172374899.92409871/v1 | This is a preprint and has not been peer-reviewed. Data may be preliminary.

Fontes, J. T., Vieira, P. E., Ekrem, T., Soares, P., & Costa, F. O. (2021). BAGS: An automated Barcode, Audit & Grade System for DNA barcode reference libraries. *Molecular Ecology Resources* ,*21* (2), 573–583. https://doi.org/10.1111/1755-0998.13262

Froese, R., & Pauly, D. (2023). *Fishbase. Available online: www.fishbase.org (accessed on 21 December 2023)*.

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning* . MIT press.

Gostel, M. R., & Kress, W. J. (2022). The expanding role of DNA barcodes: Indispensable tools for ecology, evolution, and conservation. *Diversity* , *14* (3), 213. https://doi.org/10.3390/d14030213

Haldar, C., & Nath, S. (2020). DNA barcoding for fish species identification: current status and future prospective. *Int. J. Fauna Biol. Stud* , *7* , 72–75.

Jeunen, G., Dowle, E., Edgecombe, J., von Ammon, U., Gemmell, N. J., & Cross, H. (2023). CRABS—a software program to generate curated reference databases for metabarcoding sequencing data. *Molecular Ecology Resources* , *23* (3), 725–738. https://doi.org/10.1111/1755-0998.13741

Kapoor, A., Gulli, A., Pal, S., & Chollet, F. (2022). *Deep Learning with TensorFlow and Keras: Build and deploy supervised, unsupervised, deep, and reinforcement learning models* . Packt Publishing Ltd.

Keck, F., & Altermatt, F. (2023). Management of DNA reference libraries for barcoding and metabarcoding studies with the R package refdb.*Molecular Ecology Resources* , *23* (2), 511–518. https://doi.org/10.1111/1755-0998.13723

Keck, F., Couton, M., & Altermatt, F. (2023). Navigating the seven challenges of taxonomic reference databases in metabarcoding analyses.*Molecular Ecology Resources, 23* (4), 742-755. https://doi.org/10.1111/1755-0998.13746

Kingma, D. P., & Ba, J. (2014). Adam: A method for stochastic optimization. *arXiv Preprint arXiv:1412.6980* . https://doi.org/10.48550/arXiv.1412.6980

Kirse, A., Bourlat, S. J., Langen, K., & Fonseca, V. G. (2021). Unearthing the potential of soil eDNA metabarcoding—Towards best practice advice for invertebrate biodiversity assessment.*Frontiers in Ecology and Evolution* , *9* , 337. https://doi.org/10.3389/fevo.2021.630560

Lamperti, L., Sanchez, T., Si Moussi, S., Mouillot, D., Albouy, C., Fluck, B., Bruno, M., Valentini, A., Pellissier, L., & Manel, S. (2023). New deep learning-based methods for visualizing ecosystem properties using environmental DNA metabarcoding data. *Molecular Ecology Resources* , *23* (8), 1946–1958. https://doi.org/10.1111/1755-0998.13861

Lavrador, A. S., Fontes, J. T., Vieira, P. E., Costa, F. O., & Duarte, S. (2023). Compilation, revision, and annotation of DNA barcodes of marine invertebrate non-indigenous species (NIS) occurring in European coastal regions. *Diversity* , *15* (2), 174. https://doi.org/10.3390/d15020174

Lowenstein, J. H., Amato, G., & Kolokotronis, S.-O. (2009). The real*maccoyii* : Identifying tuna sushi with DNA barcodes–contrasting characteristic attributes and genetic distances. *PloS One* ,*4* (11), e7866. https://doi.org/10.1371/journal.pone.0007866

Magoga, G., Forni, G., Brunetti, M., Meral, A., Spada, A., De Biase, A., & Montagna, M. (2022). Curation of a reference database of COI sequences for insect identification through DNA metabarcoding: COins.*Database* , *2022* , baac055. https://doi.org/10.1093/database/baac055

Meglecz, E. (2023). COInr and mkCOInr: Building and customizing a nonredundant barcoding reference database from BOLD and NCBI using a semi-automated pipeline. *Molecular Ecology Resources* ,*23* (4), 933–945. https://doi.org/10.1111/1755-0998.13756

Posted on 15 Aug 2024 | The copyright holder is the author/funder. All rights reserved. No reuse without permission. | https://doi.org/10.22541/au.172374899.92409871/v1 | This is a preprint and has not been peer-reviewed. Data may be preliminary.

Muller, J., Mitesser, O., Schaefer, H. M., Seibold, S., Busse, A., Kriegel, P., Rabl, D., Gelis, R., Arteaga, A., & Freile, J. (2023). Soundscapes and deep learning enable tracking biodiversity recovery in tropical forests. *Nature Communications* , *14* (1), 6191. https://doi.org/10.1038/s41467-023-41693-w

Nair, V., & Hinton, G. E. (2010). Rectified linear units improve restricted boltzmann machines. *Proceedings of the 27th international conference on machine learning (ICML-10),* 807–814.

Nilsson, R. H., Taylor, A. F. S., Adams, R. I., Baschien, C., Bengtsson-Palme, J., Cangren, P., Coleine, C., Daniel, H.-M., Glassman, S. I., Hirooka, Y., Irinyi, L., Iršėnaitė, R., Martin-Sanchez, P. M., Meyer, W., Oh, S.-Y., Sampaio, J. P., Seifert, K. A., Sklenář, F., Stubbe, D., … Abarenkov, K. (2018). Taxonomic annotation of public fungal ITS sequences from the built environment—A report from an April 10-11, 2017 workshop (Aberdeen, UK). *MycoKeys* ,*28* , 65–82. https://doi.org/10.3897/mycokeys.28.20887

Oliveira, L., Knebelsberger, T., Landi, M., Soares, P., Raupach, M., & Costa, F. O. (2016). Assembling and auditing a comprehensive DNA barcode reference library for European marine fishes. *Journal of Fish Biology* , *89* (6), 2741–2754. https://doi.org/10.1111/jfb.13169

Packer, L., Gibbs, J., Sheffield, C., & Hanner, R. (2009). DNA barcoding and the mediocrity of morphology. *Molecular Ecology Resources* , *9 Suppl s1* , 42–50. https://doi.org/10.1111/j.1755-0998.2009.02631.x

Pentinsaari, M., Ratnasingham, S., Miller, S. E., & Hebert, P. D. (2020). BOLD and GenBank revisited–Do identification errors arise in the lab or in the sequence libraries? *PloS One* , *15* (4), e0231814. https://doi.org/10.1371/journal.pone.0231814

Radulovici, A. E., Vieira, P. E., Duarte, S., Teixeira, M. A. L., Borges, L. M. S., Deagle, B. E., Majaneva, S., Redmond, N., Schultz, J. A., & Costa, F. O. (2021). Revision and annotation of DNA barcode records for marine invertebrates: Report of the 8th iBOL conference hackathon. *Metabarcoding and Metagenomics* , *5* , e67862. https://doi.org/10.3897/mbmg.5.67862

Raghavendra, U., Acharya, U. R., & Adeli, H. (2020). Artificial intelligence techniques for automated diagnosis of neurological disorders. *European Neurology* , *82* (1–3), 41–64. https://doi.org/10.1159/000504292

Ratnasingham, S., & Hebert, P. D. (2007). BOLD: The Barcode of Life Data System (http://www. barcodinglife.org). *Molecular Ecology Notes* , *7* (3), 355–364. https://doi.org/10.1111/j.1471-8286.2007.01678.x

Ratnasingham, S., & Hebert, P. D. (2013). A DNA-based registry for all animal species: The Barcode Index Number (BIN) system. *PloS One* ,*8* (7), e66213. https://doi.org/10.1371/journal.pone.0066213

Raybaut, P. (2009). Spyder-documentation. *Available Online at: pythonhosted.org* .

Ward, R. D., Zemlak, T. S., Innes, B. H., Last, P. R., & Hebert, P. D. N. (2005). DNA barcoding Australia's fish species. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences* , *360* (1462), 1847–1857. https://doi.org/10.1098/rstb.2005.1716

WoRMS Editorial Board. (2023). *World Register of Marine Species.* Available from http://www.marinespecies.org at VLIZ. Accessed 2023-12-21. https://doi.org/10.14284/170

Xiong, F., Shu, L., Zeng, H., Gan, X., He, S., & Peng, Z. (2022). Methodology for fish biodiversity monitoring with environmental DNA metabarcoding: The primers, databases and bioinformatic pipelines. *Water Biology and Security* , *1* (1), 100007. https://doi.org/10.1016/j.watbs.2022.100007

Yang, C.-H., Wu, K.-C., Chuang, L.-Y., & Chang, H.-W. (2021). DeepBarcoding: Deep learning for species classification using DNA barcoding. *IEEE/ACM Transactions on Computational Biology and Bioinformatics* , *19* (4), 2158–2165. https://doi.org/10.1109/TCBB.2021.3056570